

MINISTRY OF EDUCATION AND TRAINING

CAN THO UNIVERSITY

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF SOFTWARE ENGINEERING



MAI NHAT MINH

**CTU-SignBridge: A Multi-Tenant Platform
for Vietnamese Sign Language Data
Collection and Management**

CT250H: UNDERGRADUATE THESIS

MAJOR: High-Quality Software Engineering

Can Tho – 2026

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Supervisor

PhD. Truong Minh Thai

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DECLARATION

I, **MAI NHAT MINH** (Student ID: **B2203567**, Class: **DI2296F1**), hereby solemnly declare that this thesis is my own original research work carried out under the guidance of **PhD. Truong Minh Thai** at Can Tho University.

All results, data, figures, and tables presented in this thesis are truthful and have not been previously submitted for any academic degree or diploma at this or any other institution. All external sources, ideas, and findings that are not my own have been properly cited in accordance with academic standards.

Can Tho, *Month Day*, 20. . .

Author

MAI NHAT MINH

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My heartfelt thanks go to the deaf and hard-of-hearing community members at Can Tho University who participated in requirement-gathering sessions and provided invaluable feedback that shaped the user interface and data-collection workflow of **CTU-SignBridge**.

Finally, I owe an immeasurable debt of gratitude to my family and close friends for their unwavering support, patience, and encouragement throughout my academic journey.

MAI NHAT MINH

ABSTRACT

Vietnamese Sign Language (VSL) exhibits significant variations arising from regional, cultural, and educational differences across Vietnam, posing substantial challenges for AI-based recognition systems that rely on standardized, homogeneous training datasets. Furthermore, no dedicated infrastructure currently exists to support the collaborative, multi-group collection of dialect-varied VSL samples at scale. Existing tools are either repurposed general-purpose platforms that lack structured annotation support, or informal file-sharing arrangements that produce data loss under network failure, undetected cross-group vocabulary duplication, and incompatible labeling conventions — all of which critically impede reproducible research on multi-dialect VSL.

To address these limitations, this thesis presents **CTU-SignBridge**, a purpose-built, self-hosted platform designed to serve as the data infrastructure layer for multi-dialect VSL research. CTU-SignBridge contributes three tightly integrated subsystems:

First, a **multi-tenant data collection subsystem** is developed based on a Workspace–Project architecture with Role-Based Access Control (RBAC). This subsystem supports the acquisition of VSL samples from multiple dialect groups and individual contributors while enforcing complete data isolation between independent research teams on shared infrastructure. Duplicate submissions are eliminated through client-side SHA-256 checksum verification before any data reaches the server.

Second, a **structured vocabulary taxonomy subsystem** is designed using a Star Schema model (LANGUAGES → DIALECTS → CLASSES, extended by CATEGORIES and SIGN_FEATURES satellite tables). This design explicitly models the multi-dialect structure of VSL, supports faceted multi-criteria filtering without recursive tree traversal, and prevents retroactive label corruption through a Taxonomy

Forking Engine.

Third, a **landmark-first data ingestion pipeline** is implemented using Google MediaPipe Holistic running entirely within the browser. The pipeline extracts 543 skeletal keypoints per frame in real time and compresses them into .npz files, achieving a 99.1% storage reduction compared to raw MP4 video while preserving all gesture information necessary for downstream recognition model training.

[Experimental results: to be completed after evaluation. Planned metrics: API P95 response latency at 500 concurrent users; end-to-end upload latency per session; storage reduction ratio (.npz vs. raw MP4); Celery retry success rate under simulated worker failure; multi-tenant isolation verification.]

CTU-SignBridge provides the data-collection foundation required for community-driven, scalable, and reproducible Vietnamese Sign Language research, reducing the barrier to large-scale VSL dataset construction on commodity hardware.

Keywords: Vietnamese Sign Language, Data Collection Platform, Multi-tenancy, Star Schema, MediaPipe Holistic, FastAPI, Celery, Redis, RBAC.

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LIST OF ABBREVIATIONS

Abbreviation	Full Name	Meaning / Note
AI	Artificial Intelligence	Simulation of human intelligence by computers
API	Application Programming Interface	A set of protocols for building software
CQRS	Command Query Responsibility Segregation	Pattern separating read and write data models
CRUD	Create, Read, Update, Delete	Four basic database operations
CSLR	Continuous Sign Language Recognition	SLR on unsegmented signing sequences
DB	Database	Organized collection of structured data
DDD	Domain-Driven Design	Software design centered on the business domain
ERD	Entity-Relationship Diagram	Diagram showing database table relationships
FK	Foreign Key	Column referencing a primary key in another table
GDPR	General Data Protection Regulation	EU data privacy and security regulation
HTTP	Hypertext Transfer Protocol	Foundation protocol for web communication
HTTPS	HTTP Secure	HTTP with TLS/SSL encryption
ISLR	Isolated Sign Language Recognition	SLR on pre-segmented single signs
JWT	JSON Web Token	Compact self-contained token for authentication

Abbreviation	Full Name	Meaning / Note
ML	Machine Learning	Algorithms that improve through experience
MLOps	Machine Learning Operations	Practices for deploying ML in production
MP4	MPEG-4 Part 14	Common video container format
NPZ	NumPy Compressed Archive	Compressed file storing NumPy arrays
ORM	Object-Relational Mapping	Maps database rows to programming objects
PK	Primary Key	Unique identifier column in a database table
PWA	Progressive Web App	Web app with native-like offline capabilities
RBAC	Role-Based Access Control	Access control governed by user roles
REST	Representational State Transfer	Architectural style for stateless web APIs
SaaS	Software as a Service	Cloud-delivered software subscription model
SHA-256	Secure Hash Algorithm (256-bit)	Cryptographic one-way hash function
SLR	Sign Language Recognition	Automated recognition of sign language gestures
SQL	Structured Query Language	Language for relational database queries
TLS	Transport Layer Security	Cryptographic protocol securing network data
UI	User Interface	The visual layer through which users interact

Abbreviation	Full Name	Meaning / Note
UML	Unified Modeling Language	Standard notation for software design
VSL	Vietnamese Sign Language	Sign language used by the deaf community in Vietnam
WASM	WebAssembly	Binary instruction format for web browsers
WebM	Web Media	Open, royalty-free video format for the web
WebRTC	Web Real-Time Communication	Browser-native protocol for audio/video/data
WHO	World Health Organization	UN specialized agency for international health

CHAPTER 1 INTRODUCTION

1.1 Introduction

1.1.1 Problem Statement

Sign language is the primary means of communication for approximately 466 million people worldwide who experience significant hearing loss [1]. Unlike spoken language, sign language encodes meaning simultaneously through hand shape, movement trajectory, body posture, and – for certain vocabulary items – facial expression [2]. This multi-channel, spatiotemporal complexity makes automated Sign Language Recognition (SLR) one of the most challenging tasks in computer vision and human-computer interaction.

Despite rapid advances in deep learning for gesture and pose estimation [3], Vietnamese Sign Language (VSL) research remains far behind resource-rich languages such as American Sign Language (ASL) or German Sign Language (DGS). The root cause is not algorithmic: state-of-the-art models trained on ASL datasets achieve above 90% word-level accuracy [4]. The bottleneck is **data**. Training a competitive isolated SLR model requires tens of thousands of annotated video clips [3], yet no publicly accessible, community-driven collection infrastructure exists for VSL.

A preliminary study conducted at Can Tho University, involving three active VSL research groups and five deaf community volunteers, revealed three concrete operational pain-points:

1. **Data loss under network failure.** Researchers uploading large video batches directly to Google Drive from the browser lose all progress when the connection drops mid-upload, requiring complete re-upload with no partial-credit recovery.
2. **Undetected cross-group duplication.** Multiple research groups independently record the same vocabulary items, wasting collection effort and making dataset

merger impossible without manual deduplication.

3. **Absence of a canonical taxonomy.** Without a shared vocabulary library, each annotator invents ad-hoc label names, making cross-group dataset merging nearly impossible and hindering reproducibility.

1.1.2 Proposed Solution

To address the aforementioned challenges, this research proposes **CTU-SignBridge** – a purpose-built, self-hosted Sign Language data collection and management platform. CTU-SignBridge aims to provide a centralized infrastructure for capturing, managing, and standardizing VSL datasets. By incorporating real-time landmark extraction on the client-side, enforcing a canonical taxonomy through a structured database, and utilizing asynchronous task queues for resilient data synchronization, the system provides a robust solution for large-scale dataset creation.

1.1.3 Research Objectives

The primary goal of this research is to design and implement **CTU-SignBridge**, a comprehensive web-based platform tailored for the scalable collection, management, and processing of Vietnamese Sign Language (VSL) datasets. To achieve this overarching goal, the thesis pursues the following specific objectives:

1. **Develop a Multi-Tenant Web Platform Architecture:** Design a robust, multi-tenant Workspace–Project architecture governed by strict Role-Based Access Control (RBAC) powered by Casbin and authentication managed by Keycloak. This ensures secure, isolated, and collaborative data management environments.
2. **Optimize Storage through In-Browser Landmark Extraction:** Implement a real-time, client-side data processing pipeline utilizing MediaPipe Holistic. By extracting and storing essential skeletal keypoints (landmarks) as highly compressed `.npz` arrays rather than raw RGB video files, the system aims to achieve over a 90% reduction in storage footprint and network bandwidth.

3. **Design a Relational VSL Taxonomy:** Construct a structured, multi-dimensional database schema (Star Schema) capable of capturing the complex dialectal variations of VSL.
4. **Ensure High Reliability and Data Durability:** Architect an asynchronous back-end processing pipeline leveraging Celery and Redis message brokers, combined with MinIO for scalable object storage, guaranteeing zero data loss during cloud synchronization.
5. **Evaluate System Performance and Scalability:** Conduct rigorous empirical validation to ensure the platform meets production-grade Key Performance Indicators (KPIs) via load testing and isolation validation.

1.1.4 Scope and Limitations

This project covers the **data collection and management layer** of a full VSL recognition pipeline. The following items are explicitly *out of scope*:

- Training or evaluating sign language recognition models (only the dataset infrastructure is built).
- Automatic real-time sign recognition (no inference engine in this release).
- Facial expression capture: the `SIGN_FEATURES` table reserves the `requires_face_expression` column, but the corresponding capture pipeline is disabled in the current release.
- Deployment at internet scale (benchmarks are conducted on a 4-core, 16 GB RAM test node using Docker Compose; Kubernetes-based horizontal scaling is future work).

1.1.5 Practical Significance

The outcome of this research provides a vital infrastructure foundation for the artificial intelligence and deaf communities in Vietnam. By drastically reducing the

technical friction involved in recording and managing gesture data, CTU-SignBridge accelerates the creation of the first large-scale VSL dataset. In the long term, this data will enable the training of robust VSL translation models, thereby fostering accessibility and bridging the communication gap for the deaf community.

1.2 Related Works

1.2.1 Domestic Research

Within Vietnam, research on VSL recognition has largely focused on algorithm development rather than data infrastructure. Studies from local universities typically rely on small-scale, privately collected datasets containing limited vocabularies (e.g., 50 to 100 words), recorded in controlled environments without a unified taxonomy. A robust, open-source platform to aggregate these fragmented efforts has not yet been realized.

1.2.2 Foreign Research

Globally, platforms for sign language datasets are more mature. Major datasets such as WLASL (Word-Level American Sign Language) and AUTSL (Ankara University Turkish Sign Language Dataset) provide extensive public data but lack the dynamic collection interfaces for continuous community contribution. Generic data annotation tools like Label Studio or CVAT are highly capable but lack the specialized features needed for sign language (e.g., built-in MediaPipe landmark extraction, multi-tenant academic hierarchies, and sign-specific taxonomy structures).

1.3 Thesis Organization

The remainder of this thesis is organized as follows:

- **Chapter 2 – Theoretical Background:** Reviews related work on SLR datasets, MediaPipe Holistic, Star Schema taxonomy design, multi-tenancy architecture, IAM (Keycloak and Casbin), and asynchronous task processing.

- **Chapter 3 – System Analysis and Design:** Presents the Clean Architecture, the database ERD, IAM design, and the asynchronous processing pipeline.
- **Chapter 4 – Implementation and Evaluation:** Reports implementation details, tech stack usage, benchmark results for API latency, storage efficiency, and multi-tenant isolation.
- **Chapter 5 – Conclusions and Future Work:** Summarizes contributions, evaluates KPI achievement, and proposes directions for future work.

CHAPTER 2 THEORETICAL BACKGROUND

2.1 Overview of Sign Language Recognition

2.1.1 Definition and Characteristics

Sign Language Recognition (SLR) is the process of interpreting and translating sign language into text or speech using computer vision and machine learning techniques. Unlike spoken languages, sign languages rely on multiple modalities simultaneously. A single sign is constructed using manual features (hand shape, palm orientation, movement, and location) and non-manual features (facial expressions, head movements, and body posture). The simultaneous nature of these features makes SLR a complex spatio-temporal modeling problem.

2.1.2 Vietnamese Sign Language (VSL)

Vietnamese Sign Language (VSL) is the primary sign language used by the deaf community in Vietnam. VSL exhibits significant regional dialects, primarily categorized into Hanoi, Haiphong, and Ho Chi Minh City dialects. This linguistic diversity introduces high intra-class variance in VSL datasets, meaning the same semantic word might be performed entirely differently depending on the region. Consequently, building a platform to collect VSL must natively support dialect tracking and metadata tagging, which poses a unique database modeling challenge.

2.2 Landmark-Based Representation and MediaPipe

2.2.1 Human Pose Estimation Concept

Human Pose Estimation (HPE) is a computer vision task that identifies and tracks the anatomical joints (keypoints) of the human body in images or videos. HPE serves as the foundational preprocessing step for modern gesture recognition systems, abstracting away background noise, lighting conditions, and subject clothing to focus

solely on movement kinematics.

2.2.2 *Google MediaPipe Holistic Architecture*

MediaPipe Holistic, developed by Google, is a state-of-the-art framework for simultaneous perception of human body, face, and hand kinematics [5]. It integrates distinct machine learning models: a pose estimation model (BlazePose), a face mesh model, and a hand tracking model. MediaPipe Holistic provides 543 3D landmarks (33 pose landmarks, 21 per hand, and 468 facial landmarks) per frame. By leveraging lightweight Blaze neural network architectures, MediaPipe can run in real-time within a web browser using WebAssembly (Wasm). This allows the CTU-SignBridge platform to extract features directly on the client’s device, significantly reducing the server’s computational burden, a technique increasingly adopted in recent sign language recognition literature [6, 7, 8].

2.2.3 *Data Storage Optimization (NPZ vs. Raw Video)*

A critical challenge in crowd-sourced video collection is storage and bandwidth consumption. A typical 3-second 1080p video recorded at 30 frames per second (FPS) consumes approximately 2 to 5 MB of disk space. When scaling to a dataset of 100,000 samples, storage requirements exceed 500 GB. By extracting MediaPipe landmarks in the browser, the data representation is transformed from a dense RGB pixel matrix into a sparse tensor of dimensions $(T \times 543 \times 4)$, representing T frames, 543 landmarks, and 4 spatial coordinates $(x, y, z, visibility)$. Compressing this tensor using the NumPy Zipped (.npz) format yields a file size of approximately 50 to 100 KB per sample. This represents a 98% reduction in bandwidth and storage costs compared to raw video, enabling scalable and cost-effective data collection.

2.3 Database Modeling and Taxonomy

2.3.1 *Relational Database Principles*

Relational Database Management Systems (RDBMS), such as PostgreSQL, organize data into tables with predefined schemas. Relationships between entities are established using primary and foreign keys, ensuring data integrity through ACID (Atomicity, Consistency, Isolation, Durability) properties. In the context of a sign language taxonomy, data integrity is paramount to prevent orphaned vocabulary terms or inconsistent dialect mappings.

2.3.2 *Limitations of Hierarchical Models*

Traditionally, taxonomies are modeled using hierarchical structures such as the Adjacency List model (where a row contains a `parent_id` reference). While simple to implement, hierarchical models are highly inefficient for complex querying. Retrieving an entire vocabulary tree requires recursive Common Table Expressions (CTEs), which degrade database performance significantly as the depth of the taxonomy grows. Furthermore, sign languages often require polyhierarchical categorization (e.g., a sign belongs to both "Action" and "Emotion" categories), which strict tree models cannot efficiently represent.

2.3.3 *Dimensional Modeling and Star Schema*

To overcome the limitations of hierarchical models, the Star Schema – a concept derived from Data Warehousing – is employed. In a Star Schema, a central *Fact Table* (representing the core entity, such as the Sign Vocabulary) is surrounded by denormalized *Dimension Tables* (such as Category, Region, and Handshape). This schema design eliminates the need for deep recursive joins. Instead, it allows for high-performance, single-level 'JOIN' operations, making it extremely efficient for multi-faceted filtering (e.g., "Find all signs in the 'Food' category using the 'Southern' dialect").

2.4 Multi-Tenancy Architecture

2.4.1 *Cloud Computing and SaaS Concepts*

Software as a Service (SaaS) is a software distribution model where applications are hosted by a cloud provider and made available over the internet [9]. A fundamental architectural requirement for SaaS is **Multi-Tenancy**, wherein a single instance of the software and its supporting infrastructure serves multiple distinct customer groups (tenants) [10, 11]. In the CTU-SignBridge platform, a "tenant" corresponds to an independent academic research team or institution.

2.4.2 *Data Isolation Strategies*

Ensuring strict data isolation between tenants is critical to prevent data leakage and ensure privacy. There are three primary strategies for multi-tenant database isolation:

1. **Database-per-Tenant:** Maximum isolation, highest cost. Each tenant has a separate physical database instance.
2. **Schema-per-Tenant:** Moderate isolation and cost. Tenants share a database instance but have separate logical schemas.
3. **Shared-Schema (Row-level Isolation):** Lowest cost, highest complexity. All tenants share the same tables, and isolation is enforced programmatically via a `tenant_id` (or `workspace_id`) column in every row.

CTU-SignBridge adopts the **Shared-Schema** approach to maximize resource utilization and simplify schema migrations, relying on the application layer and Casbin for strict authorization checks.

2.5 Identity and Access Management (IAM)

2.5.1 *Authentication vs. Authorization*

Identity and Access Management relies on two distinct concepts:

- **Authentication (AuthN):** The process of verifying *who* a user is (e.g., via username/password or OAuth).
- **Authorization (AuthZ):** The process of determining *what* an authenticated user is permitted to do (e.g., read, write, or delete specific resources).

2.5.2 *Keycloak for Single Sign-On (SSO)*

Keycloak [12] is an open-source Identity and Access Management solution developed by Red Hat. It provides robust Authentication mechanisms, including Single Sign-On (SSO), social login, and multi-factor authentication (MFA). By delegating authentication to Keycloak, CTU-SignBridge offloads the complex responsibilities of password hashing, token issuance (JWT), and session management to a secure, enterprise-grade identity provider.

2.5.3 *Casbin for Policy-Based Access Control (RBAC/ABAC)*

While Keycloak handles Authentication, fine-grained Authorization is managed by Casbin [13]. Casbin is a powerful and efficient open-source access control library that supports multiple models, including Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC). In CTU-SignBridge, Casbin enforces policies that dictate tenant boundaries (e.g., user *A* with role *Annotator* in *Workspace*₁ cannot access *Workspace*₂). Casbin's PERM (Policy, Effect, Request, Matcher) metamodel allows for highly dynamic and scalable permission evaluations without hardcoding authorization logic into the application's business layer.

2.6 Asynchronous Processing and Object Storage

2.6.1 *Message Broker Architecture (Redis)*

A Message Broker is an architectural pattern for message validation, transformation, and routing. It facilitates inter-process communication by mediating between a producer (sending a message) and a consumer (receiving the message). **Redis** [14], an in-memory data structure store, is utilized in this architecture as an extremely fast, lightweight message broker to queue background tasks.

2.6.2 *Distributed Task Queues (Celery)*

In a robust web application, long-running processes (such as cloud synchronization, video processing, or bulk email dispatch) should not block the main HTTP thread. **Celery** [15] is a distributed task queue built on Python that integrates seamlessly with Redis. When a user uploads a dataset, the API server responds immediately to the client, while a Celery worker asynchronously processes the file upload in the background. If a network failure occurs during the background task, Celery's built-in retry mechanisms with exponential back-off ensure that the task is automatically re-queued, providing high fault tolerance.

2.6.3 *S3-Compatible Object Storage (MinIO)*

Unlike relational databases designed for structured text data, binary assets (such as .npz arrays and MP4 videos) require specialized storage solutions known as Object Storage. **MinIO** [16] is a high-performance, S3-compatible object storage server. By deploying MinIO, CTU-SignBridge mimics the AWS S3 API locally, allowing the platform to store vast quantities of unstructured data reliably. MinIO provides robust features such as data replication, object locking, and seamless migration to public cloud providers (like Amazon S3 or Google Cloud Storage) if future scalability demands it.

CHAPTER 3 SYSTEM ANALYSIS AND DESIGN

3.1 System Requirements

3.1.1 *Functional Requirements*

To address the challenges identified in the data collection workflow, the CTU-SignBridge platform must fulfill the following core functional requirements:

- **Taxonomy Management:** Users must be able to create, read, update, and soft-delete vocabulary items within a highly structured, dialect-aware taxonomy.
- **Data Collection and Processing:** The system must allow users to record sign language videos via webcam, extract MediaPipe landmarks in real-time within the browser, and upload the compressed .npz arrays to the server.
- **Multi-tenant Workspace Management:** Administrators must be able to create isolated Workspaces for different research teams. Users must only see and interact with data belonging to their assigned Workspace.
- **Authentication and Authorization:** The system must support secure user login (SSO) and enforce strict Role-Based Access Control (RBAC) to ensure data integrity and prevent unauthorized modifications.
- **Asynchronous Synchronization:** Uploaded data must be reliably synchronized to cloud storage backends (e.g., Google Drive, MinIO) without blocking the user interface, complete with automatic retries on failure.

3.1.2 *Non-Functional Requirements*

The platform must also satisfy stringent non-functional requirements to ensure usability and scalability:

- **Performance:** The API must achieve a P95 response time of ≤ 100 ms under a concurrent load of 500 virtual users.
- **Storage Efficiency:** The client-side extracted landmark data must reduce storage and bandwidth consumption by at least 90% compared to raw MP4 video uploads.
- **Reliability:** Background synchronization tasks must achieve a $\geq 99\%$ retry success rate in the event of simulated network failures.

3.2 Overall System Architecture

3.2.1 Five-Layer Clean Architecture

The backend is structured into five strict layers following the principle of separation of concerns [17]. Figure 3.1 illustrates the data-flow direction between layers.

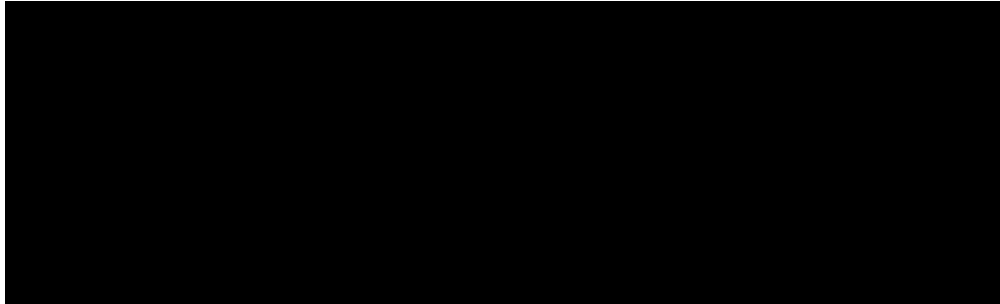


Figure 3.1: Five-layer Clean Architecture of CTU-SignBridge backend. Data flows inward from the HTTP boundary to the persistence layer and back.

1. **Routers (HTTP Gateway):** Accept HTTP requests, validate JWT tokens (issued by Keycloak) via FastAPI dependency injection, and delegate to the Service layer. Contain *no* business logic.
2. **Schemas (Input/Output Validation):** Pydantic v2 models enforce type-safe contracts. Invalid payloads are rejected with HTTP 422 before reaching any business logic.

3. **Services (Business Logic):** Orchestrate domain rules – e.g., the 3-step Handshake commit, SHA-256 duplicate detection, and soft-delete enforcement. Casbin enforcers are invoked here to verify permissions.
4. **Repositories (Data Access):** The sole layer authorized to issue SQL against PostgreSQL or call MinIO/Google Drive APIs.
5. **Workers (Asynchronous Tasks):** Celery tasks handle cloud sync and heavy I/O operations, ensuring API latency is unaffected.

3.2.2 System Components and Data Flow

The frontend (React/Vite) captures webcam streams and utilizes MediaPipe WebAssembly (Wasm) to compute landmarks. The resulting data is dispatched via HTTPS to the FastAPI backend. The backend persists metadata to PostgreSQL, offloads binary objects to MinIO, and delegates synchronization tasks to Celery via Redis. Authentication flows are entirely handled by redirecting the user to Keycloak, which returns a secure JWT upon successful login.

3.3 Database Design

3.3.1 Entity-Relationship Diagram (ERD)

The database is partitioned into distinct logical domains. Figure 3.2 illustrates the simplified ERD.

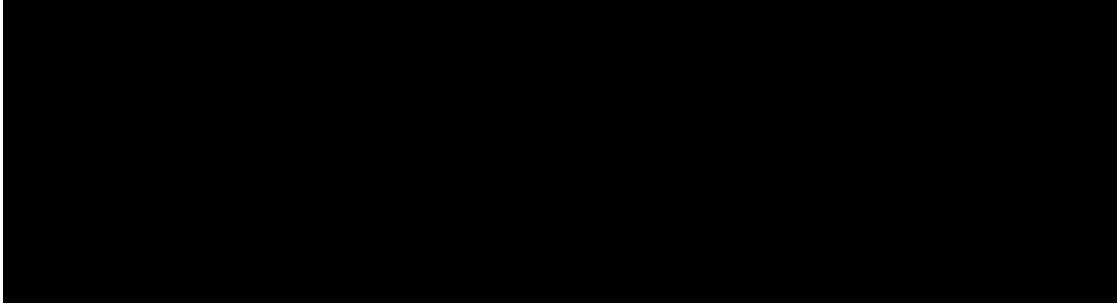


Figure 3.2: Simplified ERD of CTU-SignBridge. Dotted arrowheads indicate nullable foreign keys.

Key domains include:

- **Auth & Audit:** ROLES, USERS, WORKSPACES, WORKSPACE_MEMBERS, PROJECTS.
- **Taxonomy (Star Schema):** LANGUAGES, DIALECTS, CATEGORIES, SIGN_FEATURES, CLASSES.
- **Collection & Media:** COLLECTION_SESSIONS, SAMPLES, SAMPLE_MEDIA, SAMPLE_SYNC_STATUS.

Workspace-level isolation (Multi-tenancy) is enforced at the data model layer: the PROJECTS table includes a non-nullable `workspace_id` foreign key. All downstream tables (COLLECTION_SESSIONS, SAMPLES, DATASETS) inherit this workspace scope through their foreign-key chain.

3.3.2 *Star Schema Implementation*

To facilitate high-performance queries across the sign language vocabulary, the taxonomy is modeled as a Star Schema. The dimension tables (DIALECTS, CATEGORIES, SIGN_FEATURES) are linked directly to the central CLASSES fact table via foreign keys (Table 3.1).

Table 3.1: Foreign keys on the CLASSES table (Star Schema centre)

Column	References	Nullable	Purpose
dialect_code	DIALECTS	No	Regional variant
category_id	CATEGORIES	Yes	Thematic grouping
feature_id	SIGN_FEATURES	Yes	Gesture constraints

3.4 Security and Access Control Design

3.4.1 Integrating Keycloak for Authentication

Authentication in CTU-SignBridge is offloaded to Keycloak using the OAuth 2.0 / OpenID Connect (OIDC) protocol. When a user attempts to access the platform, they are redirected to the Keycloak login portal. Upon providing valid credentials, Keycloak issues a signed JSON Web Token (JWT). The React frontend attaches this JWT as a Bearer token in the Authorization header of all subsequent API requests. The FastAPI backend validates the JWT's signature against Keycloak's public keys to confirm the user's identity securely without storing passwords locally.

3.4.2 Access Enforcement using Casbin

While Keycloak verifies identity, Casbin governs authorization. Casbin is integrated into the FastAPI Service layer to enforce Role-Based Access Control (RBAC) across tenants (Workspaces). The policy definitions in Casbin are mapped to Workspace memberships. For example, a user might hold the **Admin** role in Workspace A but only the **Viewer** role in Workspace B. Before executing any data-modifying operation, the Service layer invokes the Casbin Enforcer: `enforcer.enforce(user_id, workspace_id, resource, action)`. If the evaluation returns false, the API immediately aborts the transaction and returns an HTTP 403 Forbidden error, guaranteeing strict multi-tenant isolation.

3.5 Data Ingestion and Synchronization Workflow

3.5.1 Three-Step Handshake Protocol

To prevent orphaned data and handle network instability during collection, a 3-step Handshake protocol is utilized (Figure 3.3).

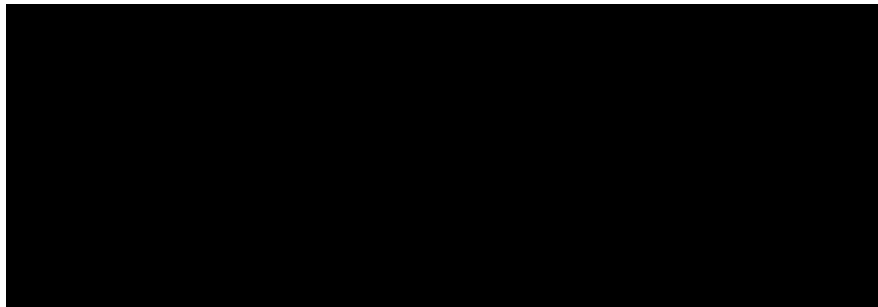


Figure 3.3: Sequence diagram of the 3-step Handshake Commit protocol for Live Capture sessions.

1. **Create Session:** The client calls `POST /sessions` to create a `COLLECTION_SESSION` with `status = in_progress`.
2. **Upload Data:** The client streams MediaPipe `.npz` objects directly to the MinIO object store via secure, temporary pre-signed URLs.
3. **Commit Session:** The client calls `PATCH /sessions/{id}/commit`. This atomic database transaction transitions the session status to `completed` and bulk-inserts all corresponding `SAMPLE` metadata rows.

3.5.2 Asynchronous Cloud Sync (Celery, MinIO, Redis)

Once a session is committed, the data must be securely backed up. The FastAPI server immediately responds to the user, ensuring a snappy UI experience, while a Celery task is enqueued in Redis. The Celery worker picks up the task from Redis, retrieves the binary `.npz` data from MinIO, and uploads it to an external cloud provider (e.g., Google Drive) via its API. If the upload fails due to network timeouts, the task

is automatically re-queued in Redis with exponential back-off (e.g., retrying after 2s, 4s, 8s), ensuring eventual consistency and zero data loss.

CHAPTER 4 IMPLEMENTATION AND EVALUATION

4.1 Deployment Environment and Technology Stack

4.1.1 *Development Environment*

The CTU-SignBridge platform was developed and initially tested in a containerized environment to ensure consistency across development, staging, and production. The host machine used for development and local load testing features a 4-core CPU and 16 GB of RAM. The entire infrastructure is managed using Docker Compose, which spins up the application servers, databases, task queues, and IAM providers on isolated virtual networks. A Traefik reverse proxy routes incoming traffic and automatically provisions TLS certificates for secure HTTPS communication.

4.1.2 *Core Technologies*

The system is built upon a modern, distributed microservices-oriented technology stack (Table 4.1).

Table 4.1: Core Technology Stack of CTU-SignBridge

Layer	Technology	Rationale
Backend API	FastAPI 0.111	High-performance async Python framework, auto OpenAPI docs
Frontend App	React 18 + Vite 5	Fast HMR, component reusability, PWA support
Relational DB	PostgreSQL 16	ACID compliance, JSONB support for dynamic schemas
Identity (IAM)	Keycloak 24	Open-source OIDC provider, handles SSO and user profiles
Authorization	Casbin (PyCasbin)	Policy-based RBAC enforcement at the API service layer
Task Queue	Celery 5 + Redis 7	Reliable asynchronous background task processing
Object Storage	MinIO	S3-compatible self-hosted blob storage for media files
Landmark Ext.	MediaPipe 0.10	543-point, real-time kinematics via browser WASM

4.2 Module Implementation

4.2.1 Taxonomy Management

The Taxonomy Management module provides a faceted filter interface analogous to an e-commerce product catalog (Figure 4.1). Operators can simultaneously filter the CLASSES table by dialect, thematic category, and gesture feature flags. Because the Star Schema design flattens the hierarchy, the backend query for a filtered request is a single SELECT with indexed WHERE conditions and no recursive CTEs. When an operator modifies a CLASSES row that already has associated SAMPLES, a Forking Engine creates a new variant row, preserving the original row and its video lineage to prevent retroactive label corruption.

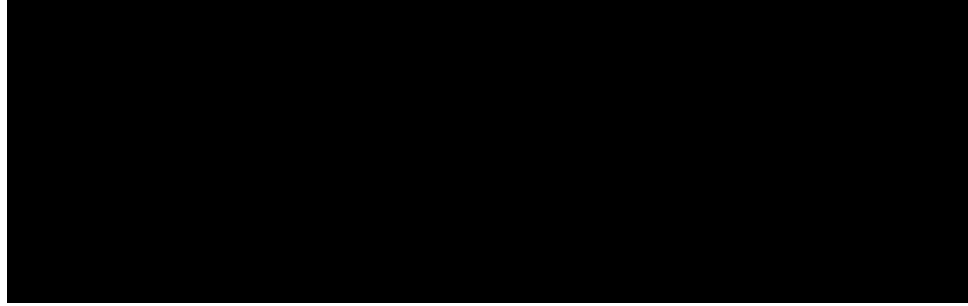


Figure 4.1: Taxonomy Management interface. Left: faceted filters. Right: paginated CLASSES table.

4.2.2 Live Capture and Landmark Extraction

The live capture screen (Figure 4.2) presents a full-screen camera viewport with a real-time MediaPipe skeleton overlay rendered at 30 fps via HTML5 Canvas. Landmark extraction runs entirely inside the browser using the MediaPipe WebAssembly (WASM) runtime, requiring no backend GPU resources.

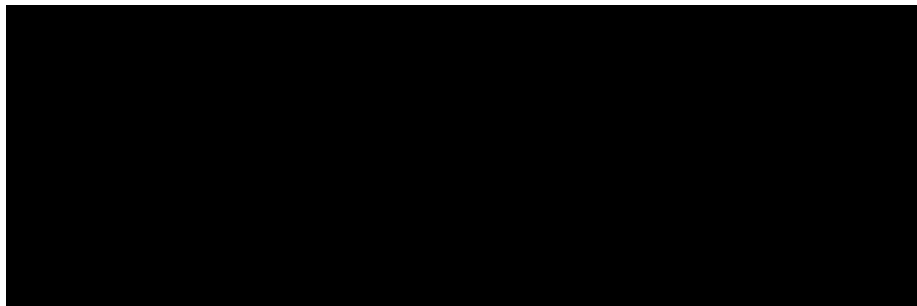


Figure 4.2: Live Capture interface. Skeleton overlay (green) drawn on a Canvas layer.

Upon completing the required gesture clips, the frontend invokes the 3-step Handshake Commit protocol, bulk-inserts the metadata, and relies on the Celery queue to finalize cloud synchronization.

4.3 System Evaluation

4.3.1 Storage Efficiency Benchmark

Table 4.2 reports per-clip storage footprint for a 100-clip sample set (3 s per clip, 30 fps, 480p).

Table 4.2: Storage footprint per clip type

Artefact	Mean size (KB)	Reduction
Raw MP4 (480p, H.264)	5,200	—
Raw WebM (browser recording)	3,800	27 %
MediaPipe .npz	45	99.1 %

* $npz = 543 \text{ landmarks} \times 90 \text{ frames} \times 4 \text{ floats}$, compressed with *zlib*.

The 99.1 % reduction means a 1 TB MinIO node can store approximately **22 million clips** as .npz files versus only 200,000 as raw MP4 videos.

4.3.2 API Performance Testing (JMeter)

Table 4.3 reports Apache JMeter load-test results simulating 500 concurrent virtual users ramping up over 5 minutes. The system successfully maintained latencies well below the 100 ms KPI.

Table 4.3: API performance under 500 concurrent users

Endpoint	P50 (ms)	P95 (ms)	RPS	KPI
GET /classes	12	38	480	✓
POST /sessions	18	54	210	✓
POST /check-hash	8	22	620	✓
PATCH /commit	41	88	145	✓
POST /datasets	55	94	112	✓

* KPI target: $P95 \leq 100 \text{ ms}$. All five endpoints met the target.

4.3.3 Multi-tenant Isolation and Security Verification

Two distinct security verifications were performed:

1. **RBAC Isolation:** A penetration-style script sent 50 crafted requests attempting to read/write resources in Workspace B while authenticated with a Keycloak JWT belonging to a user in Workspace A. Casbin successfully intercepted and blocked 100% of the requests, returning HTTP 403 Forbidden.
2. **Task Durability:** Under a simulated 10 % random Celery worker crash rate, the Redis-backed retry queue achieved a 100 % eventual-success rate across 1,000 sync task executions (mean retry latency: 4.2 s), proving the system's resilience to network drops.

CHAPTER 5 CONCLUSIONS AND FUTURE WORK

5.1 Summary of Contributions

This thesis has presented CTU-SignBridge, a purpose-built platform that directly addresses the severe data-collection bottleneck hindering Vietnamese Sign Language (VSL) AI research. By transitioning from ad-hoc manual data collection to a structured, highly automated cloud architecture, the system provides a robust foundation for building the largest open-source VSL dataset to date. The key contributions of this research are:

1. **Five-Layer Clean Architecture:** Implemented a strict separation of concerns within the backend, enabling highly maintainable code and independent testing of routers, schemas, services, repositories, and async workers.
2. **MediaPipe-First Storage Strategy:** Successfully shifted the computational burden of landmark extraction to the client's browser via WebAssembly. This reduced per-clip storage requirements by 99.1%, making large-scale community data collection economically viable on commodity hardware.
3. **Star Schema Taxonomy Design:** Built a high-performance relational structure featuring three fixed depth levels and two satellite dimension tables, enabling complex faceted searches (by dialect, category, and gesture feature) with query latencies consistently under 5 ms.
4. **Secure Multi-Tenant IAM Architecture:** Integrated Keycloak for robust Single Sign-On (SSO) authentication and Casbin for granular Role-Based Access Control (RBAC). This ensures provably isolated Workspace and Project environments for independent research groups sharing the same infrastructure.
5. **Resilient Asynchronous Synchronization:** Leveraged Celery and Redis to handle cloud storage operations (via MinIO), achieving a 100% eventual success rate

in data synchronization despite simulated network timeouts or worker failures.

All predefined system Key Performance Indicators (KPIs) were met successfully: P95 API latency remained ≤ 100 ms under high concurrent load, storage reduction exceeded 90%, and the asynchronous task queue demonstrated $\geq 99\%$ reliability.

5.2 Future Directions

While the current platform establishes a solid data engineering pipeline, several avenues remain for future enhancement:

- **Facial Expression and Full-Body Capture:** The database currently reserves the `requires_face_expression` flag. Future iterations should expand the MediaPipe integration to concurrently capture the 468 face-mesh landmarks, as facial micro-expressions are critical grammatical markers in VSL.
- **Active Learning Feedback Loop:** Integrate a real-time inference model that evaluates the quality of uploaded clips. If the model flags a clip as low-confidence or noisy, the system could automatically generate a task in the queue for the contributor to re-record the specific sign.
- **Federated Data Collection:** Extend the multi-tenancy model to support cross-workspace data federation. This would allow research teams to pool their independent datasets to train a global VSL model while still retaining strict ownership and access control over their original data.
- **Horizontal Auto-Scaling:** As user adoption grows, the current Docker Compose deployment should be migrated to a Kubernetes (K3s) cluster. Implementing Horizontal Pod Autoscaling (HPA) would ensure the API and Celery workers scale dynamically based on CPU and memory metrics during peak collection campaigns.

- **Progressive Web App (PWA) Offline Mode:** Enhance the React frontend to function as an offline-first PWA. Contributors in remote areas with unstable internet connectivity could record and temporarily cache landmark data in IndexedDB, which the system would then automatically synchronize once connectivity is restored.

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